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Multi-dimensional Gaussian Process-based Control for Compensation of Multi-state Dependent Disturbance

Hoyeong Yeo[1], Hanul Jung[2], and Sehoon Oh[1]

***Abstract*— High-precision linear motor stages have been widely used for their excellent positioning accuracy and speed. However, core-type linear motor stages have performance limitations because of various nonlinear factors including cogging force, friction, and geometrical imbalance. This paper analyzes disturbances in velocity and position domains and trains a Two-Input-Single-Output (TISO) nonlinear model using the Gaussian process for the disturbance. With this, two state-dependent disturbances are appropriately removed. As a result, the control performance with a proposed controller is enhanced. Ultimately, this paper introduces three contribution points: 1) analysis of disturbances based on position/velocity, 2) design of TISO Gaussian process model, and 3) validation of proposed controller performance through simulation.**

I. INTRODUCTION

In the industrial field, a linear motor has been achieved high control bandwidth with easy maintenance, high accuracy, and fast speed [1]. However, it is sensitive to external disturbances such as force ripple and friction [2]. Numerous studies for linear motors aim to eliminate such unpredictable nonlinear disturbances and enhance control performance.

Especially, disturbances occurring in core-type linear motors can be identified based on the geometry information of installed magnets [3]. In many studies, disturbances in the linear motor have been modeled using harmonic polynomial or multi-sinusoidal functions [4]. However, there is a challenge in adjusting the level of overfitting, which has reduced the reliability of the models. Additionally, the Gaussian process method, beneficial for modeling nonlinearity, has been utilized. While this method satisfactorily models nonlinear disturbances, it describes the relationship from a single-input-single-output (SISO) perspective [5]. Since [6] verified that disturbances in linear actuators affect position and velocity domains, there are limitations to modeling disturbances in linear motors as SISO model.

In various studies, Disturbance Observer (DOB) has been widely applied in linear motor stage due to their simple design method and high control performance. In [7], the frequency of disturbances is detected using an adaptive algorithm, and the corresponding frequency component is appropriately attenuated using a peak filter to effectively remove the target disturbances. However, there is a limitation that it only operated for a single frequency component of disturbances. Furthermore, [8] proposed a periodic adaptive DOB to learn and remove periodic disturbances. However, it is sensitive to changes in operating conditions. On the other hand, [9] designed an adaptive robust controller to compensate for the cogging force. However, an accurate cogging force model is essential. Although model-based controllers have been extensively studied to suppress disturbances in linear motors, compensating for nonlinear disturbances is still challenging.

The Gaussian process (GP) method has been applied to various control systems by learning nonlinearity based on data. Particularly, many studies have been published by combining GP and robust model-based controllers. [10] is designed with computed torque control to eliminate unexpected disturbances, while GP is utilized to compensate torque ripple with the repetitive controller by modeling disturbances at arbitrary velocity [5]. [11] proposed GP-based compensation with covariance gain to remove cogging force in linear actuators with DOB. However, [5], [10], [11] utilize GP to create SISO disturbance models. However, the issue focused on in this study is the two state dependent disturbances, which depends simultaneously on position and velocity. Therefore, it is necessary to remove two state-dependent disturbances and improve control performance using an expanded-dimensional GP.

In this paper, the proposed algorithm using a two-dimensional GP model is developed to suppress disturbances described by two states of dynamic system. Core-type linear motors suffer from unspecified nonlinear factors, including cogging force, well-known as a function of position. However, velocity-dependent disturbances practically affects the control performance, as revealed in various studies. To overcome this, a two-dimensional Gaussian process is employed to learn two state-dependent disturbances. The learned model is utilized with a robust controller, specifically a DOB, to compensate for disturbances. Furthermore, the proposed algorithm's performance is assessed by establishing a practical simulation environment, followed by comparing its control efficacy with that of existing methods.

In summary, the main contributions of this paper are as follows:

1) The characteristics of disturbances in linear motor dependent on position and velocity are experimentally verified and modeled.

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[1] H. Yeo and S. Oh are with the Department of Robotics and Mechatronics Engineering, DGIST, Daegu 42988, Korea {hoyeong23 and sehoon}@dgist.ac.kr

[2] H. Jung is with the Robotics · Mobility Research Section, Electronics and Telecommunications Research Institute (ETRI), Daegu 42994, Korea hanul@etri.re.kr

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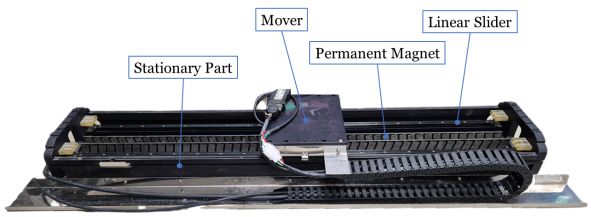

Fig. 1. Linear motor stage system from Kovery (2L-P09)

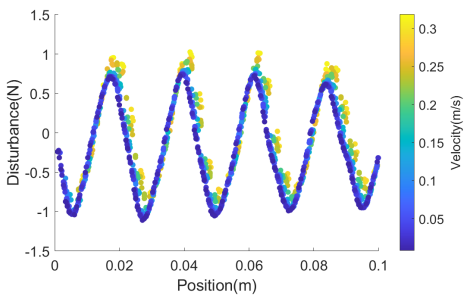

Fig. 2. Position and velocity varying disturbances of linear motor.

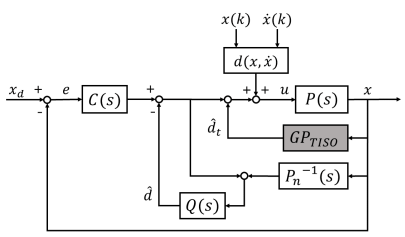

Fig. 3. Block diagram of the proposed controller including TISO GP model of disturbances.

2) A Two-dimensional Gaussian process model is designed for disturbances in the position and velocity domains.
3) A simulation setup is established to verify the effectiveness of the proposed algorithm suppressing disturbances related to both position and velocity.

II. PROBLEM STATEMENT

A. Identifying Force Ripple in Linear Motor

Disturbances generated in linear motor with powerful thrust is mainly related to the interaction between the iron core of the mover and the magnets of the stator. In iron core-type linear motors, depicted in Fig. 1, which move linearly, the mover is affect by the undefined forces due to the strong magnets uniformly distributed on the stator. The neodymium magnets on the stator are arranged at geometrically constant intervals, causing the rotor to be influenced by the magnetic field periodically depending on its position.

Fig.2 shows the disturbance information obtained when operating the linear motor at different velocities within the specific position range. This information is obtained when the control structure without GP_{TISO} is appropriately tuned as depicted in Fig.3. The estimated disturbances $\hat{d}$ are obtained using the DOB. Through Fig.2, it is evident that practical disturbances in the linear motor vary not only with position but also with velocity. In the position domain, disturbances appear as a sinusoidal wave with periodicity, while in the velocity domain, they exhibit linear bias along the position and disturbance axes. Consequently, it can be considered that actuators operating in structures with magnets positioned at equidistant intervals are influenced by position and velocity-dependent magnetic forces.

From actual acquired data, disturbances can be mathematically modeled simply in terms of position (x) and velocity ($\dot{x}$). Disturbances are a sinusoidal form in the position domain, while it shows linearity in the disturbance domain concerning velocity. This can be expressed in equations as follows,

$$F_{ripple}(x,\dot{x}) = A\sin\left(\frac{2\pi}{x_p}(x-\phi_a\dot{x}-\beta_a)\right)+\phi_b\dot{x}+\beta_b, \tag{1}$$

where A is magnitude of the signal, ϕ_a is velocity induced position bias coefficient, ϕ_b is velocity induced disturbance bias coefficient, and β_a and β_b are additional position and disturbance bias respectively.

To effectively characterize the disturbances depicted in Fig. 2, several parameters are set, where $A = 0.9$, $\phi_a = 0.005$, $\phi_b = 0.9$, $\beta_a = 0.012$, and $\beta_b = -0.2$. The mathematical disturbance model described in (1), excluding the additional bias (β_a, β_b), is employed in Section V to validate the performance of control using Multi-dimensional GP.

B. Expanded Dimension of Disturbance

In the previous study [11], disturbances are modeled solely based on position using GP, and a controller is designed to compensate for this disturbances based on the obtained model. Therefore, considering that disturbances are entirely a function of position and that unmodeled disturbances are appropriately rejected by the existing DOB, a one-dimensional GP modeling for periodic disturbances solely dependent on position posed no issues. However, the actual disturbances in the linear motor, as depicted in Fig. 2, are not perfectly periodic with respect to position, and the amplitude also varies slightly. Additionally, since bias occurs with velocity, it is not appropriate to model and compensate for disturbances based solely on information related to position.

Hence, to compensate for modeling errors arising from insufficient information, it is necessary to extend the dimension of the GP model by considering two types of information: position and velocity. While the existing position-based GP only considers uncertainty in the position axis, expanding the dimension allows for considering uncertainty in more dimensions. Among various factors influencing uncertainty of disturbance, it can be observed from Fig. 2 that position and velocity predominantly influence disturbances.

However, unlike 1-Dimensional (1D) GP, 2-Dimensional (2D) GP requires adequate information considering both position and velocity, and if sufficient disturbance information

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is available, modeling with 2D GP is feasible. The 1D GP model is a single-input-single-output GP (SISO GP) that estimates disturbances based solely on position, while the 2D GP model is a two-input-single-output GP (TISO GP) that estimates disturbances based on both position and velocity.

C. Brief Description of Proposed Controller Setting

The proposed control structure for compensating for the desired disturbance is depicted in Fig. 3. $P(s)$ represents the linear time-invariant (LTI) single-input single-output (SISO) system representing the actual linear motor, while $C(s)$ denotes the stabilizing feedback controller designed as proportional-integral-derivative (PID) control, given by,

$$C(s) = K_p + K_i \frac{1}{s} + K_d \frac{s}{\tau_d s + 1} \tag{2}$$

where K_p, K_i, and K_d are the gains of the proportional, integral, and derivative controllers, respectively, and τ_d is a time constant determining the bandwidth of the derivative controller.

GP_{TISO} refers to the proposed TISO GP model, which is elaborated in the subsequent section. The disturbance is expressed with information regarding both position and velocity, as given by (1). Additionally, there is a structure consisting of P_n^{-1} which means the inverse nominal model of $P(s)$ and a Q-filter represented by $Q(s)$ in the DOB. This nominal model utilizes the same model as [11]. Consequently, in this paper, to fully eliminate disturbance $d(x, \dot{x})$ dependent not only on position but also on velocity, a controller utilizing GP_{TISO} has been constructed, differing from previous research.

III. MULTI-DIMENSIONAL GAUSSIAN PROCESS

Generally, GP is well-known for modeling functions over distributions and is well-recognized as one of the non-parametric stochastic modeling techniques for nonlinear modeling. Compared to deterministic parametric modeling methods, algorithms utilizing GP are relatively simple but are influenced by the speed and performance of modeling depending on the number of training data points. Therefore, if the quality of the training data is good, meaning it accurately represents the characteristics of the function, it is possible to sufficiently model the function even with a small amount of data. Another factor determining the performance of GP is selecting an appropriate kernel based on the characteristics of the data. By choosing the kernel properly, it is possible to adjust the desired properties of the system model, such as periodicity or smoothness.

A. Gaussian Process for Multi-Dimensional Disturbance

In this study, disturbance considering both position and velocity is modeled using GP, and it is utilized in the proposed controller shown in Fig. 3 to compensate for the disturbance. GP is commonly employed in many studies to estimate unknown output using observations in one dimension. But, the system to be applied in this study requires consideration of both position and velocity. Therefore, it is necessary to extend the dimension of GP to two dimensions.

The modeled disturbance($d(X)$) is dependent on both position and velocity, as given by (1), and observation(d_n) is obtained in practical environments with measurement noise. The state vector $X = \begin{bmatrix} x & \dot{x} \end{bmatrix}^\top \in \mathbb{R}^{2\times 1}$ determining the disturbance includes position (x) and velocity ($\dot{x}$). Therefore, the observation is as follows,

$$d_n(X) = d(X) + \epsilon, \text{ with } \epsilon \sim \mathcal{N}(0, \sigma_n^2), \tag{3}$$

where $d(X)$ is the true disturbance affected by position and velocity, ϵ is the measurement noise, which is independent identically distributed zero-mean Gaussian noise with variance σ_n^2.

To estimate the disturbances accurately using observations with noise, a joint Gaussian distribution is employed. The set of N states and disturbances used for disturbance estimation is defined as follows,

$$\boldsymbol{X} = \begin{bmatrix} X_1 & X_2 & \cdots & X_N \end{bmatrix}^\top, \tag{4a}$$

$$\boldsymbol{d}_n = \begin{bmatrix} d_{n,1} & d_{n,2} & \cdots & d_{n,N} \end{bmatrix}^\top, \tag{4b}$$

The joint Gaussian distribution of the new disturbance data ($\tilde{\boldsymbol{d}}_*$) for M test states $\boldsymbol{X}_* \in \mathbb{R}^M$, including state and disturbance observations $\mathcal{D}(\boldsymbol{X}, \boldsymbol{d}_n)$, is as follows,

$$\begin{bmatrix} \boldsymbol{d}_n \\ \tilde{\boldsymbol{d}}_* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K + \sigma_n^2 I_N & K_* \\ K_*^\top & K_{**} \end{bmatrix} \right), \tag{5}$$

where $K \in \mathbb{R}^{N\times N}$ is the kernel matrix of train data $k(\boldsymbol{X}, \boldsymbol{X})$, and similar for $K_* = k(\boldsymbol{X}, \boldsymbol{X}_*) \in \mathbb{R}^{N\times M}$ and $K_{**} = k(\boldsymbol{X}_*, \boldsymbol{X}_*) \in \mathbb{R}^{M\times M}$ are kernel matrices, with N the number of observations and M the number of test points.

The posterior distribution of $\tilde{\boldsymbol{d}}_*$ for the test data $\boldsymbol{X}_*$ is given by,

$$p(\tilde{\boldsymbol{d}}_* | \mathcal{D}, \boldsymbol{X}_*) = \mathcal{N}(\mu_d, \Sigma_d), \tag{6}$$

where μ_d and Σ_d are the posterior mean and covariance [12], respectively, as follows,

$$\mu_d = K_*^\top (K + \sigma_n^2 I_N)^{-1} \boldsymbol{d}_n, \tag{7a}$$

$$\Sigma_d = K_{**} - K_*^\top (K + \sigma_n^2 I_N)^{-1} K_*, \tag{7b}$$

where $I_N \in \mathbb{R}^{N\times N}$ is an identity matrix. The unobserved disturbance affecting to the system can be estimated by (7a).

The proposed controller ensures high control performance by appropriately utilizing (7a). For the real-time implementation of GP control, the estimated disturbance $\hat{d}_*$ for a single test data X_* is compensated at each sampling instant by computing it as the posterior mean. However, the process of obtaining the inverse matrix in (7a) requires considerable computational resources. Therefore, to alleviate this, after obtaining the train data, α is precomputed for GP prediction and hold it as follows,

$$\alpha = (K + \sigma_n^2 I_N)^{-1} \boldsymbol{d}_n, \tag{8a}$$

$$\hat{d}_t = K_*^\top \alpha, \tag{8b}$$

$K_*^\top$ in (8b) requires computation based solely on the training data $\boldsymbol{X}$ and a single real-time measurement point $X_* \in \mathbb{R}^{2\times 1}$. Additionally, due to the characteristics of the estimated disturbance in this study, which allow for minimizing

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the amount of training data, the computational burden does not pose a significant issue for system implementation. Consequently, it is possible to obtain the estimated 2D disturbances in real-time using (8b).

B. Multi-Dimensional Kernel for Gaussian Process

The selection of an appropriate kernel function can significantly affect the efficiency and performance of GP regression. In this paper, to train a TISO GP model for the disturbance $d_n \sim \mathcal{GP}(0, k(X_p, X_q))$, we will utilize the square exponential kernel to form a smooth model by referencing neighboring data. $k(X_p, X_q)$ is the kernel function between one state (X_p) and another state (X_q), following the expression,

$$k(X_p, X_q) = \sigma_f^2 \exp(-\frac{1}{2}\bar{X}^\top L^{-1}\bar{X}) \tag{9}$$

where $\bar{X} = (X_p - X_q)$ and $L = \text{diag}(\ell_1^2, \ell_2^2)$. Unlike the 1D kernel, there are two length scales (ℓ_1, ℓ_2), which means setting reference lengths for each dimension of the data. The length scale, acting as a low-pass filter, determines the smoothness of the result [13]. A larger value results in a smoother GP model, while a smaller value includes high-frequency components in the output.

ℓ_1 has a smaller value compared to ℓ_2, indicating that the characteristics of the position data are considered more complex than the velocity data, and closer measured data points should be referenced. With a larger value for ℓ_2, it implies that, due to the less complex characteristics of the velocity data adopted in this study, data points further away can also be referenced to achieve a smoother modeling. σ_f is signal variance and is used to scale the uncertainty of the disturbance. The hyperparameters θ for setting the kernel function (9) can be defined as follows,

$$\theta = \begin{bmatrix} \ell_1 & \ell_2 & \sigma_f \end{bmatrix} \tag{10}$$

Since the setting of the hyperparameters θ significantly affects the performance of GP regression, it is crucial to select them appropriately. While it is possible to manually adjust the hyperparameters while checking GP results, it is more effective to use methods that can automatically tune them. One prominent method is to optimize the likelihood, which quantifies the suitability of the distribution for the given data, as the objective function. To effectively optimize it, the marginal log-likelihood equation is used,

$$\log p(\boldsymbol{d}_n|X) = -\frac{1}{2}\boldsymbol{d}_n^\top K_n^{-1}\boldsymbol{d}_n - \frac{1}{2}\log|K_n| - \frac{n}{2}\log 2\pi \tag{11}$$

where $K_n = K + \sigma_n^2 I_N$. In this study, the hyperparameters θ of kernel can be obtained by maximizing (11) using a gradient-based optimization method, resulting in the optimal hyperparameters $\theta^\star$ [12].

C. Procedure of Training and Compensation

To compensate for disturbances based on the trained GP model for states, Procedure 1 is followed. Through this process, a TISO GP considering position and velocity can be trained, and disturbances can be compensated for in real-time. However, it is necessary to consider the possibility that real-time performance can not be maintained due to computational load. As the number of data used for training increases, the computational burden dominated also increases, which can affect real-time compensation. The number of training data required for GP training varies depending on the environment and should be appropriately determined as it also affects the regression performance.

Procedure 1 : TISO GP compensation design

1) **Obtain $\mathcal{D}$ for train GP.**
   a) Set range of data to be obtained.
   b) Obtain $\boldsymbol{d}_n$ for the determined range.
2) **Train a TISO GP model using all $\mathcal{D}(\boldsymbol{X}, \boldsymbol{d}_n)$ obtained in step i) using** (8a)**.**
   a) Set the hyperparameters θ of kernel
3) **Compensate for the disturbances for each new sample** (X_*) **at each step using** (8b)**.**

IV. SIMULATION CASE STUDY

A. Simulation Environment Setting

The simulation configuration is depicted in Fig. 3. $P(s)$ is set as a second-order mass-damper system,

$$P(s) = \frac{1}{Js^2 + Bs} \tag{12}$$

where inertia $J = 0.238 \cdot 10^{-3}\text{kg} \cdot \text{m}^2$, damping $B = 3.810 \cdot 10^{-3}\text{kg} \cdot \text{m}^2/\text{s}$, and $C(s)$ which is designed as given by,

$$C(s) = 22.5591 + 472.4766\frac{1}{s} + 0.1496s \tag{13}$$

The gains of PID controller are tuned manually. Additionally, to prevent the causality problem in D controller, a cut-off frequency of Q-filter is also set as 100Hz. $C(s)$ is discretized by bilinear transform with sampling frequency $f_s = 1000$Hz.

The disturbance used in the simulation is described by,

$$d(x, \dot{x}) = A\sin(\frac{2\pi}{x_p}(x - \phi_a\dot{x})) + \phi_b\dot{x} \tag{14}$$

where $A = 0.9$, $x_p = 0.022$m, $\phi_a = 0.05$, and $\phi_b = 0.9$ which effectively characterizes disturbances of the linear motor as shown in Fig. 2. Therefore, the disturbance in (14) is non-periodic in the time domain but periodic in the position domain under the simulation reference as shown in Fig. 4.

B. The Region for Data Acquisition

To conduct the simulation, it is necessary to identify the motion range of the system and then obtain data to apply to train the GP model. The reason is that the performance of GP is unreliable if data does not exist near the state to be estimated. The operating range of the system for the reference in Fig. 4 corresponds to Fig. 5. The range of positions is 0∼0.3m, and the range of velocities is 0∼0.08m/s.

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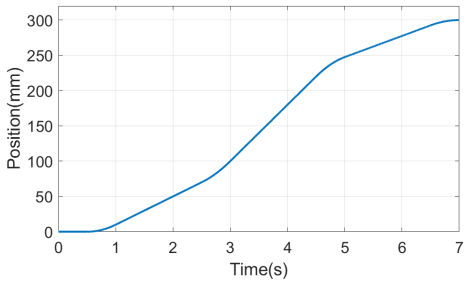

Fig. 4. Position reference used in the simulation

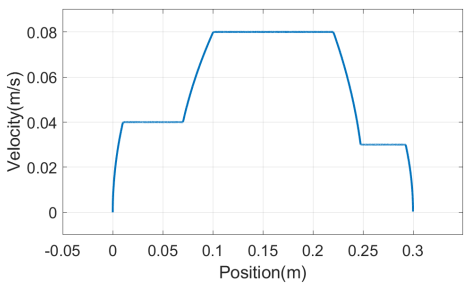

Fig. 5. The region of position and velocity reference

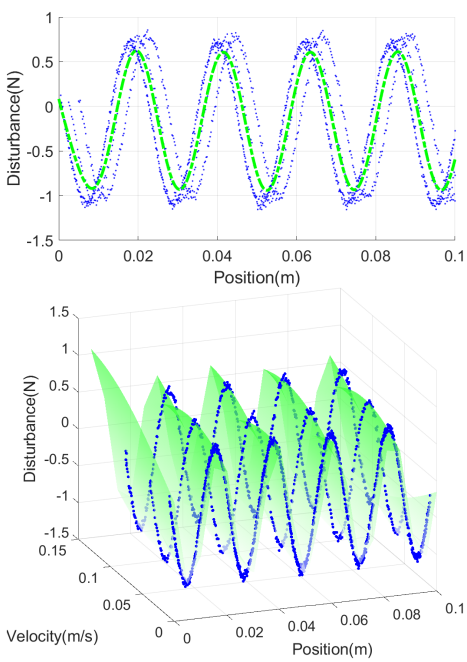

Fig. 6. Regression results of GP trained with 1D (Top) and 2D (Bottom) kernels. Actual disturbances based on each dimension of the state (—), and regression results of SISO(- -), TISO(■) GP trained based on actual disturbances.

To ensure the reliability of TISO GP implementation, it is essential for the data acquisition range to contain the operating range. Therefore the data acquisition range is set to positions $0{\sim}0.4$m and velocities $0{\sim}0.1$m/s. To obtain data within this range, position control is conducted at constant speeds, which are 0.01, 0.05, and 0.1 m/s, as shown in Fig. 6.

C. Hyperparameter Setting of TISO GP kernel

Example 1. 1D kernel induced disturbance model

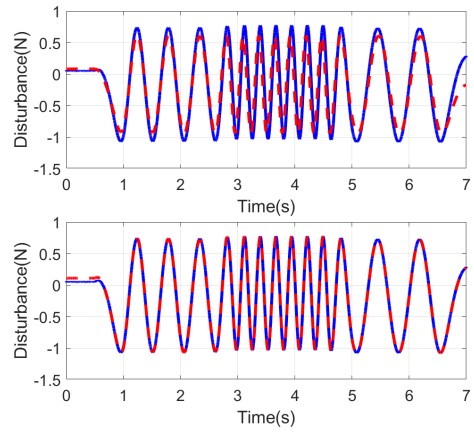

Fig. 7. Estimation of disturbance by SISO (Top) and TISO (Bottom) GP in time domain. Actual disturbances (—), estimated disturbances by GP (- -).

To compare the effectiveness, 1D kernel case is set to obtain the results. Since its performance is shown based on 1D data, the state X includes only the position information x. Therefore, the hyperparameters, in this case, are defined as $\ell_{1\mathrm{D}}$ and $\sigma_{f,1\mathrm{D}}$, as follows,

$$\theta_{1\mathrm{D}} = \begin{bmatrix} \ell_{1\mathrm{D}} & \sigma_{f,1\mathrm{D}} \end{bmatrix} \tag{15}$$

The optimal hyperparameters obtained by optimizing the marginal log likelihood are as follows: $\ell_{1\mathrm{D}} = 0.0095$ and $\sigma_{f,1\mathrm{D}} = 1.5417$. Using the 1D kernel defined with these hyperparameters, the SISO GP regression results are shown in the top plot of Fig. 6. Since the magnitude of disturbances varies with the velocity for a given position, it cannot accurately represent all observations. Therefore, the regression results represent values closer to the average of multiple disturbances for a given position.

Example 2. 2D kernel induced disturbance model

The TISO GP regression method proposed in this paper considers both position and velocity to learn the disturbance model compared to SISO GP. Thus, the optimal hyperparameters are $\ell_1 = 0.0058$, $\ell_2 = 0.148$, and $\sigma_f = 2.4633$. The TISO GP regression results are shown in the bottom plot of Fig. 6. It can cover variations in disturbances with respect to velocity. In the bottom plot of Fig. 6, although there is no data in the velocity range of $0.05{\sim}0.1$m/s, the TISO GP estimates for disturbances data which can be obtained using the posterior mean based on nearby data.

Additionally, the larger value of ℓ_2 compared to ℓ_1 means that the information in the velocity domain is less complex than in the position domain. Since the velocity information is relatively monotonic, the length scale is set to a larger value.

The results of estimating disturbances using the GP model are shown in Fig. 7, and it is observed that the estimation performance varies depending on the GP dimension. The estimation performance of the SISO GP is poor in the region of high velocity ($3 \sim 4.5$s), which is attributed to the model being trained without considering velocity.

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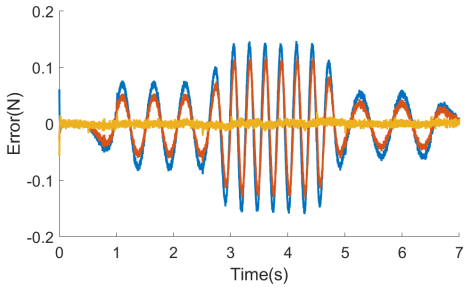

Fig. 8. Comparison of disturbance estimation errors over time. Only DOB(—), DOB+SISO GP(—), and DOB+TISO GP(—)

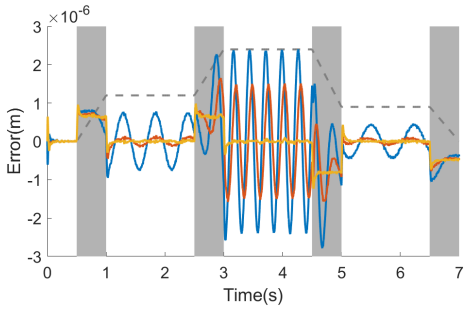

Fig. 9. Comparison of position errors over time. Only DOB(—), DOB+SISO GP(—), DOB+TISO GP(—), arbitrarily scaled velocity reference(- -), and acceleration area(■)

D. Comparison Result

In this section, the effect of setting the dimension of GP is demonstrated through the simulation for compensating disturbances. Fig. 8 shows the disturbance estimation performance of only DOB, which is without GP component, the SISO GP, which considers only position, and the TISO GP, which considers position and velocity. By comparing the errors, the effectiveness of the compensation is verified. It is observed that the only DOB case, which estimates disturbances in the time domain, performs the worst, and the SISO GP shows a large estimation error compared with the TISO GP. Both the DOB and SISO GP rely on a single domain, either time or position. However, the disturbances considered in this study significantly affect velocity changes. In particular, in only DOB and SISO GP cases, the performances decrease most in the interval of highest velocity, $3{\sim}4.5$s.

Fig. 9 illustrates the effectiveness of disturbance rejection of the proposed controller through error comparison. While all controllers show similar performance during acceleration, the controllers with GP show better performance while velocity is constant compared to the only DOB case. Particularly, the TISO GP-based controller shows the best performance in the fast range.

V. CONCLUSION AND FUTURE WORK

In this paper, the proposed method eliminates the disturbances influenced by two states. Therefore, the TISO GP model is designed to estimate the two-state dependent disturbances and apply the GP model to the controller. The proposed method effectively removed disturbances varying in position and velocity, which is verified in a practical simulation environment compared to the conventional methods.

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